# AI/ML Overview



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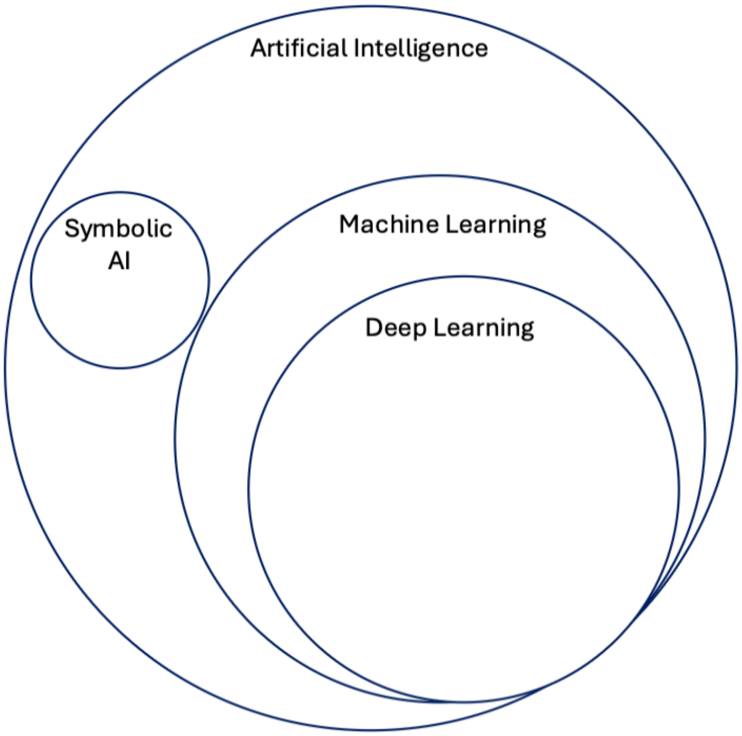
# Concepts in Artificial Intelligence

The goal of this page is to define and explain common terms associated within the field of artificial intelligence (AI) and more specifically machine learning (ML). As Figure 1, below, shows AI is a broad field which contains multiple sub-fields such as machine learning.




## Artificial Intelligence (AI)

The concept of artificial intelligence or "AI" was developed around the 1950s and can be concisely defined as **the effort to automate intellectual tasks normally performed by humans**.[1] AI is the general field that encompasses machine learning and deep learning. However, AI is not limited to learning methods, but extends to hardcoded rules as well, this approach is known as *symbolic AI* and was the dominant paradigm from 1950s until about the 1980s.[1]

## Machine Learning (ML)

Machine learning, a branch of AI, arises from a general set of questions such as "could a computer go beyond what we *order* it to perform and learn on its own to perform?" Essentially, instead of inputting rules and data to generate an answer, ML achieves to take in as an input data and answers to generate rules, which can in turn be applied to new data.[1] Thus, a ML system is trained instead if programmed. ML began to flourish in the 1990s, but really took off as the main field within AI as larger dataset and faster hardware became available.

## Deep Learning (DL)

DL is a sub-field within ML, which learns data representations by emphasising learning about that data over successive layers of increasingly meaningful representations.[1]

## MLOps

Machine learning operations refers to a set of practices that help in the development, refrainment, and deployment of ML models. Machine learning generally begins with data curation and preparation whereby data may originate at multiple, varying sources within an organization or from a customer. Data is reviewed, aggregated, cleaned and feature engineered. An exploratory phase may be involved, which requires experimentation with various model architectures and types, leading to multiple model versions, data versions, and experiment tracking recording the specific of the hyperparameter details along with the specific data and model characteristics. After experimentation, models are refined and optimized in order to address a set of business or research problems. MLOps implementations help to systemically and simultaneously manage the development and release of the ML models as code and data changes. An optimal MLOps system treats ML assets (data, experiments, code) similar to other continuous integration and deliver environments. The deployment of models ought to be standardized with the ability to run alongside the applications and services that are normally consumed in release process.

## References

[1] Chollet, Francois. Deep Learning with Python (2018).

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# Fundamentals of Machine Learning

There are four general branches of machine learning (ML).

## Supervised learning

The most common case of ML involves learning to map input data to target or label data, often called annotations. **Self-supervised learning** is a specific type of supervised learning without the need for human-annotations.

### Regression

Regression models predict a numeric value in a continuous set.

### Classification

Classification models predict the likelihood that an item belongs to a specific, discrete category.

## Unsupervised learning

This type of learning involves finding interesting transformations of the input data without any target or annotations.

### Clustering

A type of learning without any correct labels, which groups together examples in the data set based on intrinsic data characterizations.

## Generative AI

Is a class of models that creates content from user input. For example, generative AI can create unique images, music compositions, and jokes; it can summarize articles, explain how to perform a task, or edit a photo. Generative model are described as "type of input" to "type of output", some examples include:

1. Text-to-text
2. Text-to-image
3. Text-to-video
4. Image-to-text

## Reinforcement learning

This type of learning involves an *agent* that receives information about its environment and learns to choose actions that will maximize some reward.

# Model Evaluation

Evaluating a model general concerns splitting available data into three sets: training data, validation data and test data. General, a model is trained or directly learns from training data and is validated or evaluated with validation data as part of the training process. Final, a held-out set or test set of data is used to obtain final metrics for model performance.

## Model Inference

Inference occurs after a model as been sufficiently training on a data cohort and evaluated on a set of performance metrics. Once the performance metrics are met, a model is able to performance a predication or inference, to unlabelled data.

## References

[1] Chollet, Francois. Deep Learning with Python (2018).

[2] https://developers.google.com/machine-learning/glossary.

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# AI/ML Glossary

A comprehensive glossary of common terms used in Artificial Intelligence and Machine Learning. Navigate by clicking on letters below or scroll to browse all terms. Definitions are referenced from industry-standard sources[1][2].

Quick Navigation - Jump to Letter

A B C D E F G I L M N O P R S T U V

## A

### Accuracy

The fraction of predictions that a classification model got correct. For binary classification, accuracy is calculated as: (True Positives + True Negatives) / (Total Examples).

METRICS

### Activation Function

A function that enables neural networks to learn nonlinear (complex) relationships between features and the label. Common activation functions include ReLU, sigmoid, and tanh.

NEURAL NETWORKS

### Agent

Software that can reason about user inputs to plan and execute actions on behalf of the user. In reinforcement learning, the entity that uses a policy to maximize expected return gained from interacting with an environment.

AI SYSTEMS

### Artificial General Intelligence (AGI)

A non-human mechanism that demonstrates a broad range of problem-solving, creativity, and adaptability across diverse domains, similar to human cognitive abilities.

AI CONCEPTS

### Artificial Intelligence (AI)

The effort to automate intellectual tasks normally performed by humans. AI encompasses machine learning, deep learning, and rule-based symbolic AI systems.

AI CONCEPTS

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B

## B

### Backpropagation

The primary algorithm for performing gradient descent on neural networks. It calculates the gradient of the loss function with respect to the weights of the network.

TRAINING

### Batch

The set of examples used in one iteration of model training. Batch size refers to the number of examples in a batch.

TRAINING

### Bias

An intercept or offset from an origin. In machine learning, bias allows models to shift the activation function to better fit the data. Also refers to unfair prejudice in model predictions.

MODEL PARAMETERS

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## C C

### Classification

A type of supervised learning where the model predicts which category (or class) an input belongs to. For example, determining if an email is spam or not spam.

ML TASKS

### Clustering

An unsupervised learning technique that groups similar examples together without predefined labels. The algorithm discovers natural groupings in the data.

UNSUPERVISED LEARNING

### Convolutional Neural Network (CNN)

A type of neural network specialized for processing grid-like data such as images. CNNs use convolutional layers to automatically learn spatial hierarchies of features.

NEURAL NETWORKS

### Cross-Validation

A technique for assessing how well a model generalizes to independent data by partitioning the original dataset into training and validation subsets multiple times.

EVALUATION

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## D

### Dataset

A collection of examples used for training, validation, or testing a machine learning model. Each example consists of features and (optionally) a label.

DATA

### Deep Learning (DL)

A subfield of machine learning that uses neural networks with multiple layers (deep neural networks) to learn hierarchical representations of data.

ML TECHNIQUES

### Dropout

A regularization technique where randomly selected neurons are ignored during training, helping to prevent overfitting by reducing co-adaptation of neurons.

REGULARIZATION

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## E

### Embedding

A mapping from discrete objects (such as words) to vectors of continuous numbers. Embeddings capture semantic relationships between objects in a lower-dimensional space.

REPRESENTATION

### Ensemble Learning

A technique that combines predictions from multiple models to create a more robust and accurate final prediction than any individual model.

ML TECHNIQUES

### Epoch

One complete pass through the entire training dataset. Training typically involves multiple epochs to allow the model to learn patterns in the data.

TRAINING

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F

## F

### Feature

An input variable used in making predictions. In a dataset about houses, features might include the number of bedrooms, square footage, or location.

DATA

### Feature Engineering

The process of creating new features or transforming existing features to make machine learning algorithms work better. This can include normalization, encoding, and creating interaction terms.

DATA PREPARATION

### Fine-tuning

The process of adapting a pre-trained model to a specific task or dataset by continuing training on task-specific data with a smaller learning rate.

TRANSFER LEARNING

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G

## G

### Generative AI

A class of AI models that creates new content (text, images, music, code) from user input. Examples include text-to-text, text-to-image, and text-to-video models.

AI SYSTEMS

### Gradient Descent

An optimization algorithm that iteratively adjusts model parameters to minimize the loss function by moving in the direction of steepest descent of the gradient.

OPTIMIZATION

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## I I

### Inference

The process of using a trained model to make predictions on new, unseen data. This occurs after the model has been trained and validated.

DEPLOYMENT

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## L L

### Label

The "answer" or target value in supervised learning. For example, in a spam classifier, the label indicates whether an email is spam or not spam.

SUPERVISED LEARNING

### Large Language Model (LLM)

A Transformer-based language model trained on vast amounts of text data, such as GPT, Claude, or Gemini. More formally, it estimates the probability of tokens or sequences of tokens.

NLP

### Learning Rate

A hyperparameter that controls how much to adjust model parameters during each training step. Too high can cause unstable training; too low can make training extremely slow.

HYPERPARAMETERS

### Loss Function

A function that measures how far a model's predictions are from the actual labels. During training, the goal is to minimize this loss. Common examples include cross-entropy and mean squared error.

TRAINING

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## M M

### Machine Learning (ML)

A branch of AI where computers learn patterns from data without being explicitly programmed. Instead of coding rules, ML systems learn rules from examples of inputs and outputs.

AI CONCEPTS

### MLOps

Machine Learning Operations - a set of practices that combines ML, DevOps, and data engineering to deploy and maintain ML systems reliably and efficiently in production.

OPERATIONS

### Model

A mathematical representation of a process learned from data. In machine learning, a model defines the relationship between features and predictions.

CORE CONCEPTS

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N

## N

### Neural Network

A machine learning model inspired by the structure of the human brain, consisting of interconnected layers of nodes (neurons) that process and transform input data.

NEURAL NETWORKS

### Normalization

The process of scaling features to a standard range (often 0 to 1 or -1 to 1) to improve training stability and speed. Common techniques include min-max scaling and z-score normalization.

DATA PREPARATION

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## O O

### Overfitting

When a model learns the training data too well, including its noise and outliers, resulting in poor performance on new data. The model has memorized rather than generalized.

MODEL ISSUES

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## P P

### Precision

A classification metric that answers: "Of all the examples the model predicted as positive, what percentage were actually positive?" Calculated as: True Positives / (True Positives + False Positives).

METRICS

### Pre-training

The initial phase of training a model on a large general dataset before fine-tuning it on a specific task. Common in transfer learning and large language models.

TRANSFER LEARNING

### Prompt

The input text given to a language model to generate a response. Prompt engineering involves crafting effective prompts to get desired outputs from LLMs.

NLP

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## R

### Recall

A classification metric that answers: "Of all the actual positive examples, what percentage did the model correctly identify?" Calculated as: True Positives / (True Positives + False Negatives).

METRICS

### Recurrent Neural Network (RNN)

A type of neural network designed for sequential data, where connections between nodes form a directed cycle, allowing information to persist across time steps.

NEURAL NETWORKS

### Regression

A type of supervised learning where the model predicts a continuous numeric value. For example, predicting house prices or temperature.

ML TASKS

### Regularization

Techniques used to prevent overfitting by adding constraints or penalties to the model. Common methods include L1 regularization, L2 regularization, and dropout.

REGULARIZATION

### Reinforcement Learning

A type of learning where an agent learns to make decisions by interacting with an environment, receiving rewards or penalties, and learning to maximize cumulative reward.

ML TECHNIQUES

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## S S

### Self-Supervised Learning

A type of supervised learning that doesn't require human-labeled data. The model creates its own labels from the data structure (e.g., predicting masked words in a sentence).

ML TECHNIQUES

### Supervised Learning

The most common type of machine learning, where models learn to map input features to known output labels using labeled training data.

ML TECHNIQUES

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T

## T

### Token

The atomic unit used for model training and inference in NLP. A token can be a word, character, or subword. Tokens are the basic units language models use for processing text.

NLP

### Training Data

The subset of the dataset used to train the model. The model learns patterns and relationships from this data during the training process.

DATA

### Transfer Learning

A technique where a model trained on one task is adapted for a different but related task, leveraging the knowledge gained from the original training.

ML TECHNIQUES

### Transformer

A neural network architecture that uses attention mechanisms to process sequential data in parallel, forming the foundation of modern large language models like GPT and BERT.

NEURAL NETWORKS

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## U U

### Underfitting

When a model is too simple to capture the underlying patterns in the data, resulting in poor performance on both training and test data.

MODEL ISSUES

### Unsupervised Learning

A type of machine learning that finds patterns in data without labeled outputs. Common techniques include clustering, dimensionality reduction, and anomaly detection.

ML TECHNIQUES

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## V V

### Validation Data

A subset of data used to evaluate the model during training and tune hyperparameters. It helps prevent overfitting by providing feedback on generalization performance.

DATA

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## References


[1] https://developers.google.com/machine-learning/glossary

[2] Chollet, Francois. Deep Learning with Python (2018)


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